

ADVANCED PASSWORD CRACKER

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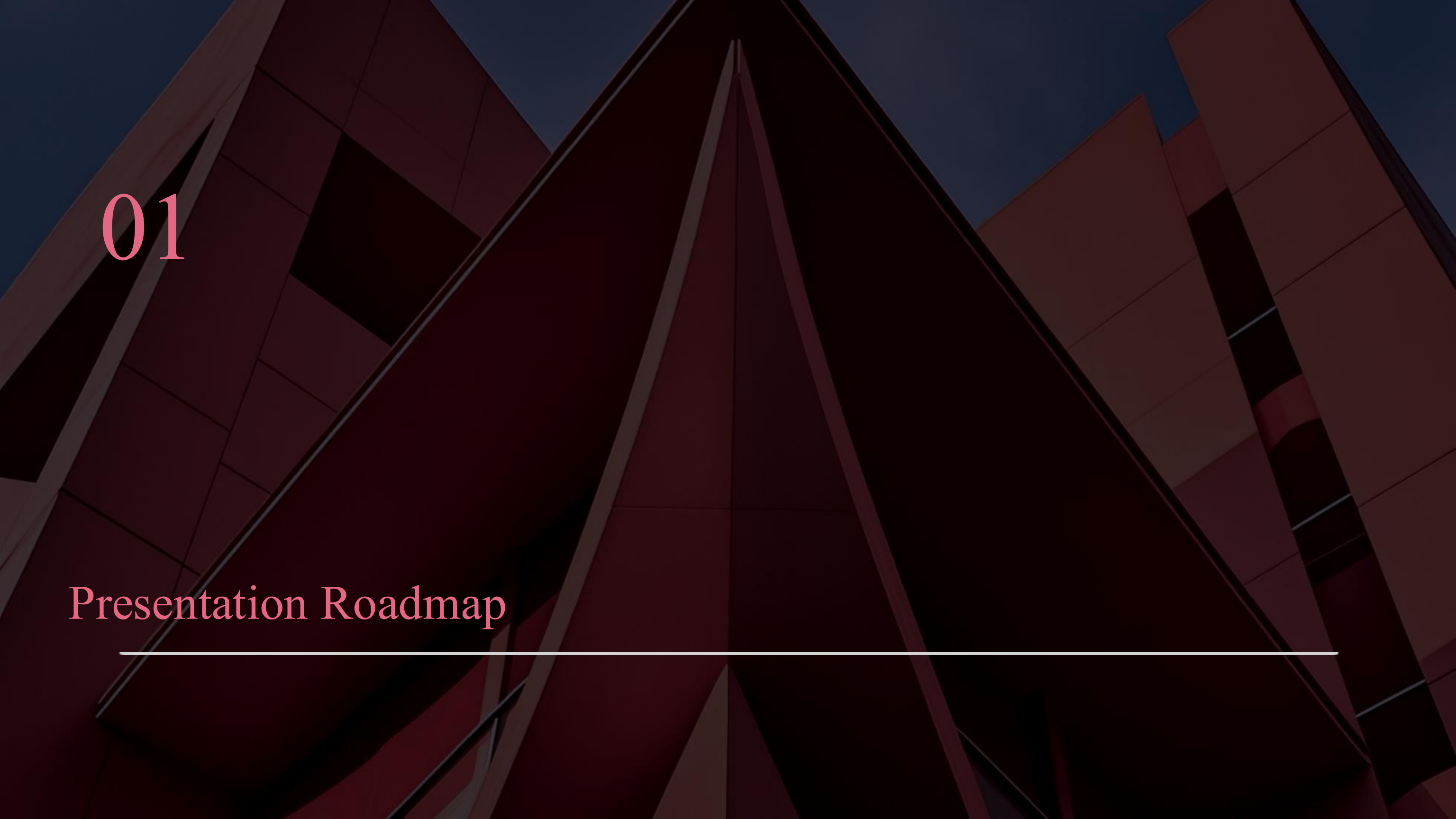
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01

Presentation Roadmap

What We'll Cover Today

In this presentation, we will guide you through the essential aspects of our project.

Project Overview: Introduction to the Password Cracking machine.

Core Features: High-performance GPUs and distributed cracking architecture.

Technical Architecture: Deep dive into hardware specs and the software platform.

Security Mechanisms: The technical safeguards we've implemented to ensure user safety.

Challenges & Solutions: Insights on how we addressed key development hurdles.

Use Cases & Strategic Benefits: Exploring the real-world value of this project.

The Team & Company: Introduction to Supraja Technologies and the developers behind this initiative.

Summary & Next Steps: Conclusion and future vision for the project.

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Project Overview



Enhancing password Privacy with Python Security

The Problem

Many systems remain exposed because users choose weak or reused passwords and organizations lack accurate, prioritized visibility into credential weaknesses.

Our Solution

Build an **authorized** security-auditing tool that simulates attacker behaviors at a high level to identify weak or compromised credentials and then delivers prioritized remediation steps, stronger password policies, and secure storage recommendations.



Key Objectives

- 1 Password-protected cracking control to restrict unauthorized access.
- 2 Advanced intrusion detection with evidence capture for accountability.
- 3 System-wide password blocking capability to enhance security.
- 4 Secure logging and immediate email notifications for any unauthorized access attempts.

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Core Features

Secure Password Control & Intrusion Detection

Our system boasts a variety of features designed to safeguard user privacy effectively.

- 1  **Authentication-Based Access:** Only authorized users can access the password analysis dashboard using a secure password, preventing unauthorized viewing of sensitive logs.
- 2  **Password Monitoring:** Captures and records all password typed on the system, providing a detailed view of user activity for analysis.
- 3  **SHA-256 Hashing of Sensitive Data:** Any sensitive data, like passwords detected in password, is securely hashed before storage to protect user privacy..
- 4  **Email Alerts with Reports:** Sends instant email notifications with logs or suspicious activity reports when abnormal behavior is detected.
- 5  **Activity Logging & Analysis:** Maintains timestamped logs of all password and system events, allowing detailed review and detection of potential threats.

The background of the slide is a photograph of a modern building with a distinctive red, vertically-slatted facade. The building has several large windows, some of which are visible. In the background, a cityscape is visible under a clear sky, with various buildings and hills in the distance. The overall tone of the image is somewhat muted, with the red of the building providing a strong contrast.

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Technical Architecture

System Components & Integration

To understand the functioning of our solution, we present the architecture and components involved.

1 Frontend

Built with Python 3.8+ using Tkinter to create a responsive graphical user interface.

2 Database

SQLite is utilized for securely storing hashed passwords and maintaining activity logs.

3 Password Operations

Python libraries such as pynput are employed to monitor and capture password activity efficiently and in real-time.

4 Alerts

The system can generate logs and send email alerts via SMTP when suspicious activity or specific key patterns are detected.

5 System Control

Admin-level controls are implemented to start, pause, or stop password activities, ensuring privacy and authorized access only.



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Implementation Workflow

From User Request to System Action

Understanding the workflow helps in visualizing how the system operates.

- 1 User requests Password access.
- 2 The system prompts for password authentication.
- 3 The entered password is verified against the SHA-256 hash stored in the database.
- 4 If Success: Password access is granted to the user.
- 5 If Failure: Intrusion detection triggers video capture to document the attempt.
- 6 An email alert with evidence is sent to the user or admin.
- 7 All activity is logged in the database for future reference.
- 8 The system can proactively block password access at the OS level to prevent unauthorized usage.

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Key Security Mechanisms

Our Multi-Layered Defense Strategy

To ensure comprehensive protection, we have implemented several security mechanisms.

Mechanism	Description	Benefit
Password Auth	Protected access control	Prevents unauthorized use of the camera.
SHA-256 Hashing	Cryptographic hashing of passwords	Secures credentials against data breaches.
Intrusion Detection	Captures video evidence on failed attempts	Provides crucial evidence for alerts and forensic analysis.
Email Alerts	SMTP notifications with evidence	Ensures that users are immediately informed of any intrusions.
Activity Logging	Records all access attempts	Supports auditing and analysis of system usag e.



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Challenges & Solutions

Addressing Development Hurdles

Throughout the development process, we encountered several challenges and devised effective solutions.

1 Challenge

Password Complexity

Long and complex passwords drastically slow down brute-force or dictionary attacks, often making advanced cracking feats computationally expensive or unfeasible.

2 Challenge

Defensive Mechanisms

Account lockouts and rate-limiting features restrict repeated password attempts, forcing attackers to use slower strategies or risk early detection.

3 Challenge

Hashing and Salting

Modern password storage with strong hashing and unique salt values render rainbow table attacks ineffective, complicating direct lookup methods.

4 Challenge

Distributed Cracking Coordination

Running large-scale attacks across multiple devices demands advanced orchestration and can trigger defensive network monitoring, reducing overall success rates.

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Performance & Security Metrics

Optimized for Responsiveness and Insight

Our system's performance is crucial for ensuring a seamless user experience while maintaining security.



Optimized for Responsiveness and Insight

Left Column (Performance Metrics)

- Async Operations: Ensures a smooth, non-blocking user interface.
- Cached Status Checks: Reduces redundant processing and improves efficiency.
- Modern UI Design: Responsive design ensures functionality across various devices.
- Safe Resource Cleanup: Prevents potential system instability that could arise from improper resource management.

Right Column (Metrics to be Collected)

- Intrusion Attempts: Number of unauthorized access attempts detected by the system.
- Email Alerts Sent: Total number of notifications dispatched to users.
- Successful Authentications: Count of valid access grants logged by the system.
- Logs Created: Total number of activity records stored in the database for auditing.j.,

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Use Cases

Use cases:-

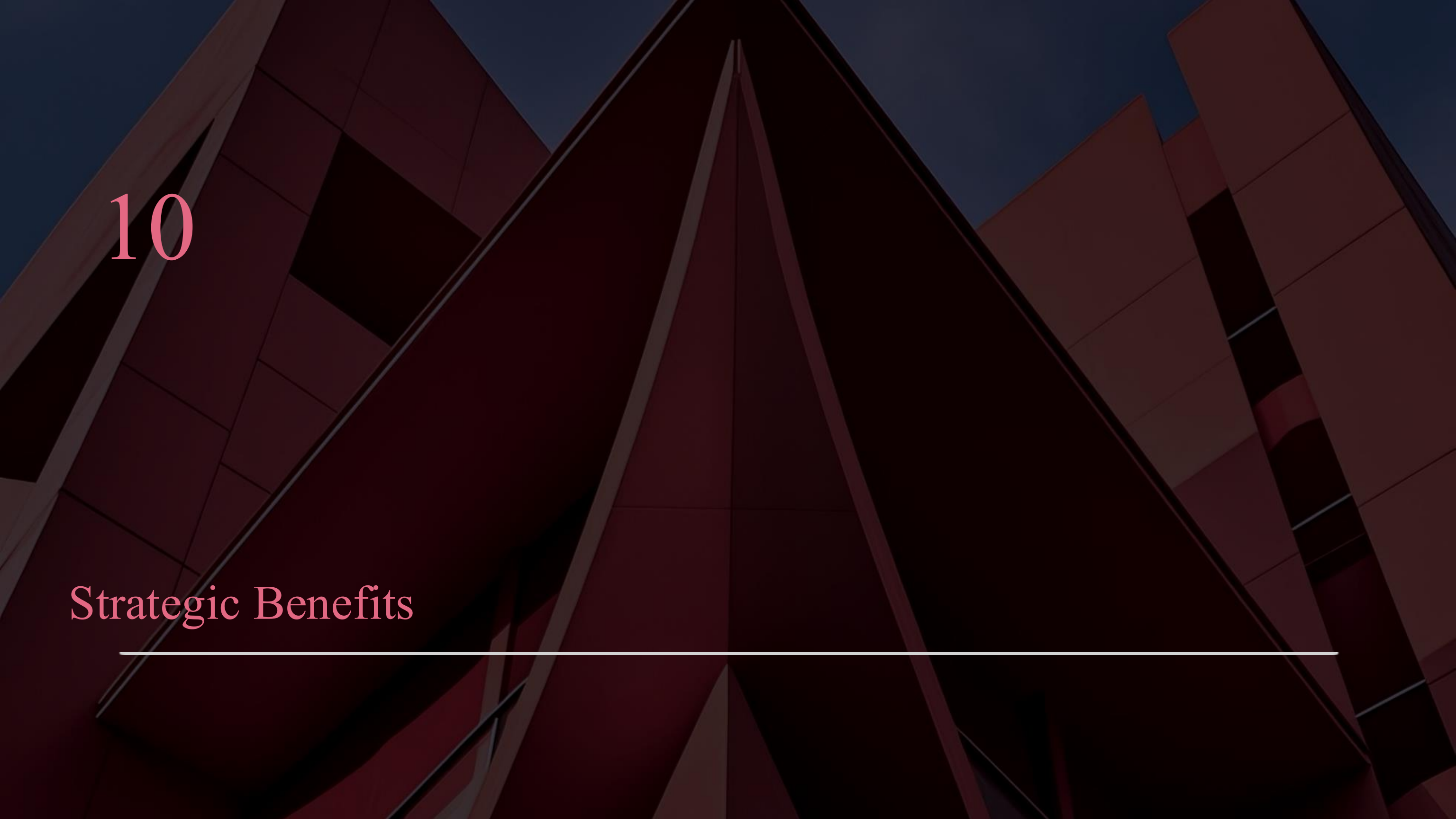
Our Camera Privacy Manager has numerous applications in enhancing IT security across various domains.

Penetrating testing/security:ethical hacking
test system password strenght identify
weak accounts before attackers

Real time project:simulate real world attacks on
organizations

Penetration test scoping (educational, authorized
and testing

Incident response triage to validate breach impact (with permission).



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Strategic Benefits

Why This Project Matters

The importance of our Camera Privacy Manager extends beyond just technology—it has significant implications for user privacy and security.

Enhances User Privacy: Directly protects against unauthorized surveillance and ensures user peace of mind.

Improves Security Posture: Adds a crucial layer of defense against evolving cyber threats.

Provides Actionable Alerts: Immediate notifications with video evidence drastically enhance response times during security breaches.

Supports Compliance: Serves as a tangible tool to help organizations meet stringent privacy regulations.

Demonstrates Python Innovation: Showcases the combination of modern programming techniques with deep system-level controls.

Builds Cybersecurity Expertise: Fosters practical skills in security development for interns involved in the project.

Developer Team

Snehitha: Focused on authentication and user interface modules.

Pragna Sri: Specialized in intrusion detection and activity logging.

Pravasya: Handled database integration and password hashing mechanisms.

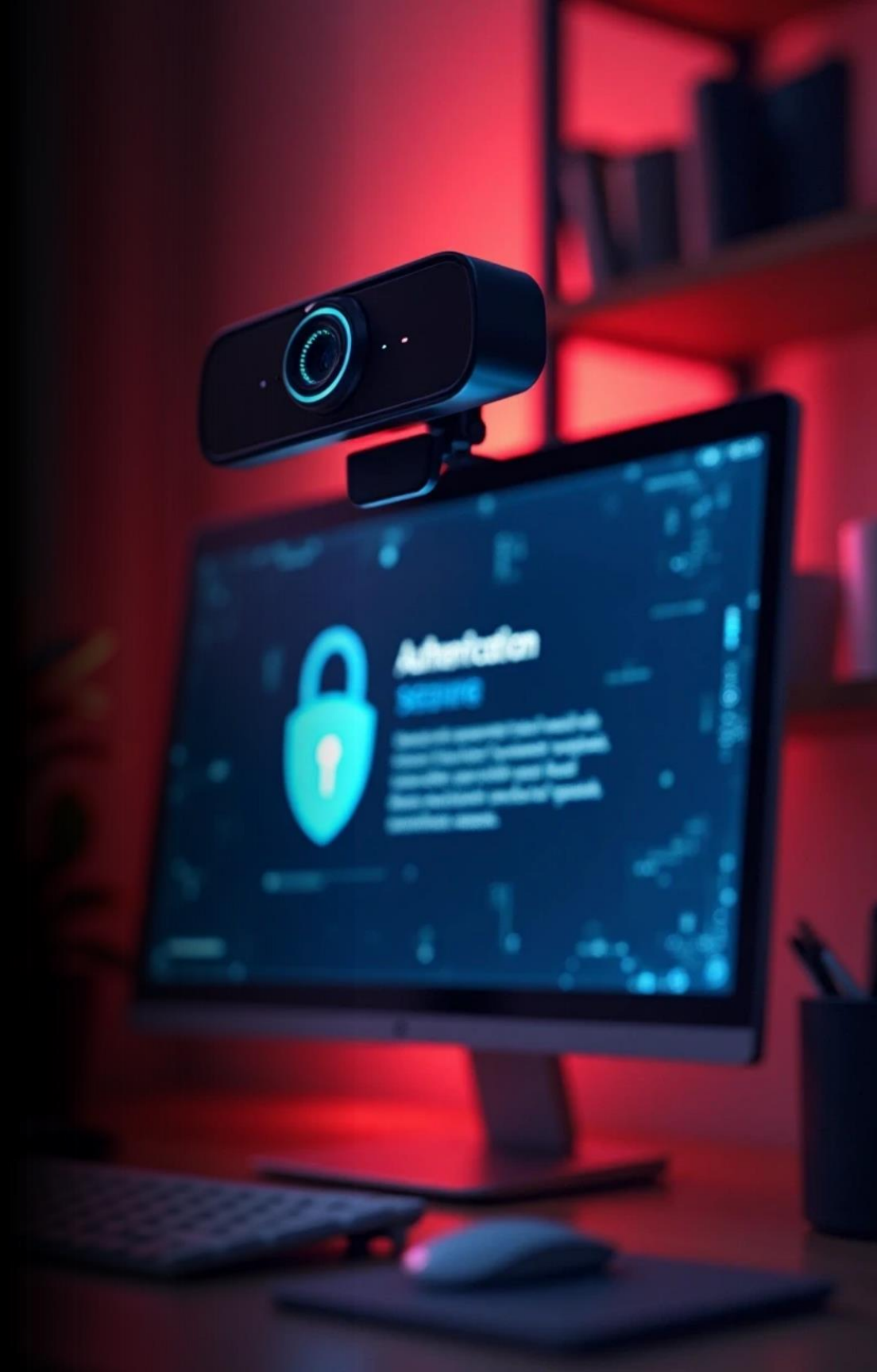


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Summary & Next Steps

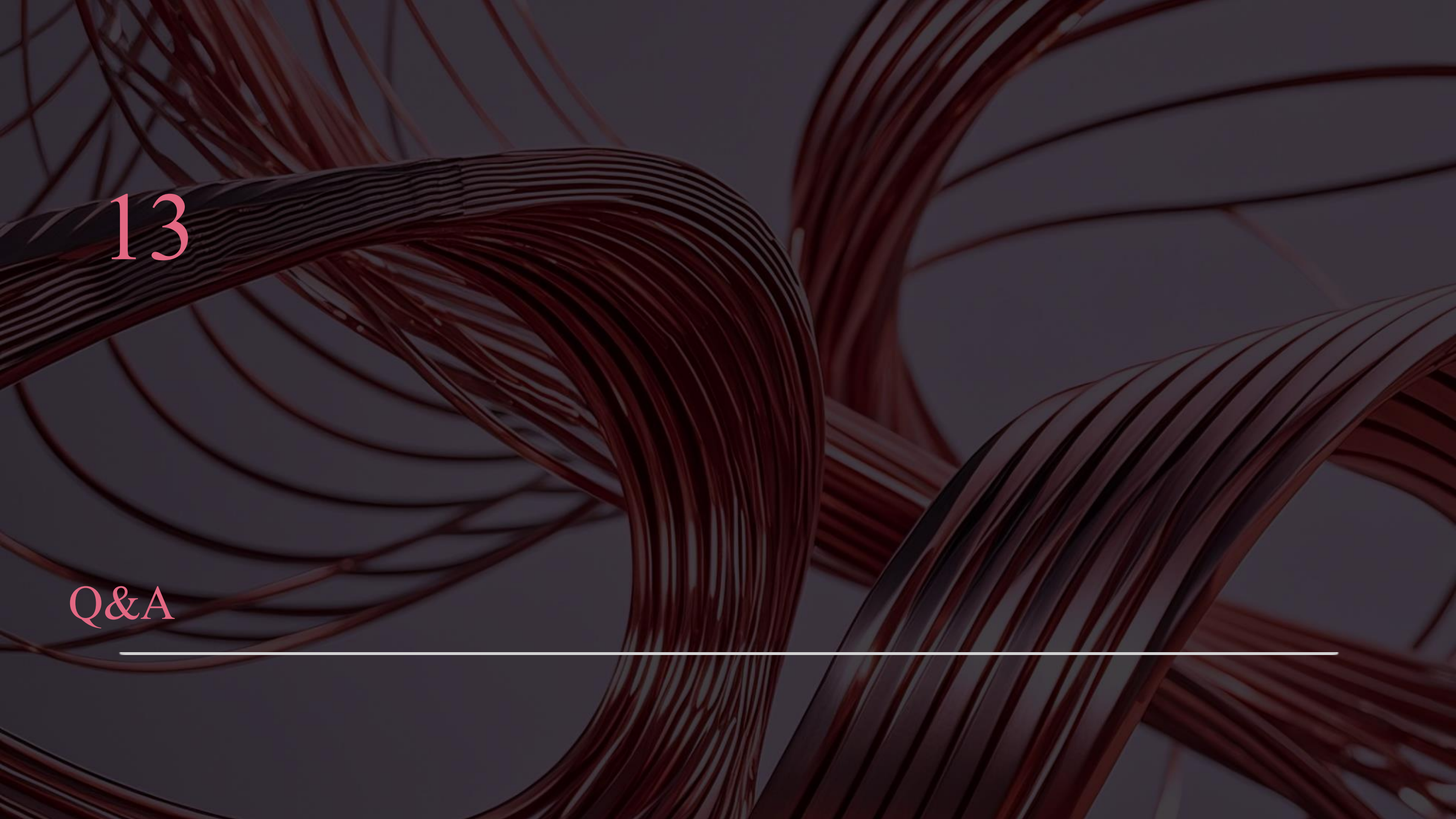
Strengthening Privacy in the Digital Age

This is an educational demonstration of password cracking concepts. This helps in privacy strengthening.



Next Steps

- 1 Add Argon2/bcrypt checks, integrate limited GPU support, and improve wordlist management
- 2 Operationalize: Add logging, consent checks, and exportable reports for audits
- 3 Expand the system's compatibility beyond Windows to reach a wider user base.
- 4 Ethics: Ensure all testing is authorized and compliant with laws and policies
- 5 Continue to drive internship-led innovation to address emerging cybersecurity threats.



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Q&A



We appreciate your time and attention.

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Questions? We welcome any questions you might have regarding our project.



Thank You